

Synopsis of
Mini Project On



“Medi Mind Using Naive Bayesian Algorithm For Multiple Disease”

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Abstract

MediMind is a MERN stack-based system for multi-disease health analysis. It works for heart disease, diabetes, breast cancer, and lung cancer. The system uses datasets like Heart Disease dataset, PIMA Diabetes dataset, Breast Cancer dataset, and Lung Cancer X-ray/CT dataset. Different algorithms are applied: Logistic Regression, Decision Tree, and Naive Bayes. Users can log in, upload prescriptions or medical images, and download custom PDF reports. MongoDB is used to store user details and analysis records, and real-time notifications are given for easy use..

Keywords:

- MERN Stack
- Multi-disease health analysis
- Heart disease prediction
- Diabetes prediction
- Breast cancer detection
- Lung cancer detection
- Medical datasets
- Heart Disease dataset
- PIMA Diabetes dataset
- Breast Cancer dataset
- Lung Cancer X-ray/CT dataset
- Machine learning algorithms

Chapter 1

INTRODUCTION

Chapter 1: Introduction

1. In recent years, healthcare has undergone a rapid transformation with the rise of digital technologies. However, many individuals still face challenges in getting timely medical diagnosis and analysis due to limited access to healthcare professionals and diagnostic facilities. Early detection of diseases such as heart disease, diabetes, breast cancer, and lung cancer plays a vital role in saving lives and improving treatment outcomes.
2. Traditional medical diagnosis often involves physical consultations and manual report evaluations, which are time-consuming and sometimes prone to error. With the advancement of machine learning and artificial intelligence (AI), it is now possible to automate disease prediction and provide accurate health analysis using digital medical reports and datasets.
3. The proposed system, MediMind, is a web-based platform designed to detect and predict diseases like heart disease, diabetes, breast cancer, and lung cancer using machine learning algorithms. It enables users to securely upload medical prescriptions, reports, or images (such as X-rays or CT scans), and receive instant health analysis along with a customized PDF report.
4. The system utilizes well-known medical datasets such as Heart Disease Dataset, PIMA Diabetes Dataset, Breast Cancer Dataset, and Lung Cancer X-ray/CT scans, ensuring reliable and accurate disease prediction results.
5. MediMind uses algorithms such as Logistic Regression, Decision Tree, and Naïve Bayes to provide efficient disease analysis and diagnosis. The system also ensures user security and privacy by using MongoDB for storing medical data, allowing easy access to past health reports while maintaining confidentiality.
6. The project aims to provide an accessible and secure platform that empowers individuals to monitor their health regularly, detect potential diseases early, and take necessary preventive measures in time. MediMind bridges the gap between patients and technology, offering quick and accurate medical insights through machine learning.

1.1 Motivation

- ❖ Many people today experience various health issues but do not get timely diagnosis because healthcare facilities are either unavailable, expensive, or difficult to access. In many cases, diseases such as heart disease, diabetes, and cancer are diagnosed at later stages when treatment becomes more complex and less effective.
- ❖ Early detection of these diseases can save lives, reduce healthcare costs, and improve the quality of life. With the growth of AI-based health systems, it has become possible to detect diseases at an early stage through data-driven analysis. However, most available systems are either too complex or not user-friendly for ordinary users.

1.2 Objectives

- ❖ To detect multiple diseases such as Heart Disease, Diabetes, Breast Cancer, and Lung Cancer using advanced machine learning algorithms.
- ❖ To allow users to securely upload medical reports or images and receive instant automated health analysis.
- ❖ To ensure data security and privacy by storing user records in MongoDB with encrypted access.
- ❖ To develop a MERN stack-based web platform for reliable and scalable performance.
- ❖ To provide users with personalized diagnostic reports in a simple and easy-to-understand format.
- ❖ To help users take timely preventive measures by detecting diseases early.

1.3 Scope

The scope of MediMind focuses on building an intelligent and user-friendly health analysis system that predicts multiple diseases with high accuracy using machine learning techniques.

This system will:

- Analyze user-provided data such as X-rays, CT scans, or medical test results and generate instant diagnostic reports.
- Support prediction for multiple diseases including Heart Disease, Diabetes, Breast Cancer, and Lung Cancer using reliable medical datasets.
- Utilize MongoDB to store and retrieve past user health records, ensuring data security and easy access.
- Provide automated health reports that help users understand their health conditions without needing manual interpretation.
- Be scalable and flexible to include more diseases in the future with additional datasets and improved models.

The main intention behind developing MediMind is to assist users in detecting diseases early, helping them take preventive measures before the condition becomes critical. This platform is not a substitute for professional medical consultation, but an intelligent support system that enables users to monitor their health efficiently and make informed decisions.

Chapter 2

Literature Review

Chapter 2: Literature Review

2.1 Related Work

A data-driven approach to predicting diabetes and cardiovascular disease with machine learning (Miertschin et al., 2019) uses survey and laboratory data from the NHANES dataset to train supervised ML models (logistic regression, SVM, random forest, gradient boosting) for diabetes and cardiovascular disease. They achieved an AU-ROC of ~ 0.86 for diabetes and ~ 0.83 for CVD using full data. [BioMed Central](#)

Deep Neural Network Based Ensemble learning Algorithms for the healthcare system (diagnosis of chronic diseases) (Abdollahi et al., 2021) review and implement ensemble methods (neural-network-based) for chronic-disease diagnosis (including diabetes, heart disease, cancer). They report very high accuracies (98–100%) in specific cases. [arXiv](#)

Med-BERT: pre-trained contextualized embeddings on large-scale structured electronic health records for disease prediction (Rasmy et al., 2020) introduce a BERT-style pre-training approach on ~ 28 million EHR entries and fine-tune for heart failure in diabetic patients and pancreatic cancer — showing that pre-training improves prediction, especially with limited data. [arXiv](#)

Novel Deep Learning Architecture for Heart Disease Prediction using Convolutional Neural Network (Hussain et al., 2021) propose a 1-D convolutional neural network architecture for heart-disease classification using clinical parameters, reporting $\sim 96\%$ test accuracy on their dataset. [arXiv](#)

Cardiovascular Health Management in Diabetic Patients with Machine-Learning-Driven Predictions and Interventions (Jose et al., 2024) examine ML-based prediction and intervention strategies for cardiovascular health in diabetic patients, showing the synergy of multi-disease modelling (diabetes + CVD) in improving outcomes. [MDPI](#)

Relevance to Our Project

- These works show the feasibility and power of machine learning for early detection of diseases relevant to our platform (heart disease, diabetes, cancer).
- They highlight that multi-disease prediction is gaining traction (e.g., diabetes + heart) — which aligns with our project that covers multiple diseases (heart, diabetes, breast cancer, lung cancer).
- They show that using large datasets, advanced architectures (e.g., CNN, BERT), and ensemble methods can improve accuracy — guiding our choice of algorithms (Logistic Regression, Decision Tree, Naïve Bayes) and potential future enhancements.
- They show current gaps: some focus only on one disease; some require large EHR datasets; some still do not integrate user-uploaded medical images (X-rays/CT) or multi-modal inputs. This gives us a space to contribute by combining multi-disease prediction + user-friendly upload + image/report input + integrated ML + secure dat

2.2 Existing System

In the existing healthcare environment, the process of disease detection and diagnosis is primarily manual and time-consuming. Most healthcare institutions rely on traditional diagnostic procedures, where medical professionals interpret clinical reports, X-rays, CT scans, and test results to determine the likelihood of diseases such as heart disease, diabetes, breast cancer, or lung cancer.

Although hospitals and laboratories have adopted digital record-keeping systems, these systems generally lack intelligent analysis capabilities. They are used merely for data storage and report generation, not for predicting or analyzing a patient's health condition automatically. Patients often have to wait for specialists to review results, which can lead to delays in early diagnosis and treatment—a critical issue for life-threatening diseases.

Some existing online tools attempt to predict certain diseases using limited parameters such as blood pressure, glucose level, and BMI. However, these systems have major limitations:

1. **Single-Disease Focus** – Most available tools are designed to predict only one disease (e.g., diabetes or heart disease). They do not offer multi-disease analysis within a single platform.
2. **Lack of Integration** – Existing systems do not combine data from multiple sources such as medical images, reports, and patient history.
3. **Low Accuracy and Outdated Models** – Many systems still use simple statistical models or linear regression techniques that cannot handle complex non-linear relationships in medical data.
4. **No Secure Data Handling** – Patient information is often stored without strong encryption or privacy mechanisms, posing serious security concerns.
5. **Poor User Interface** – Most systems are designed for healthcare professionals rather than for general users, making them difficult for patients to use independently.

Because of these limitations, current systems fail to provide comprehensive, accurate, and secure early disease detection. There is a strong need for an intelligent, automated, and user-friendly platform that can analyze multiple diseases using machine learning models while ensuring data privacy.

Chapter 3

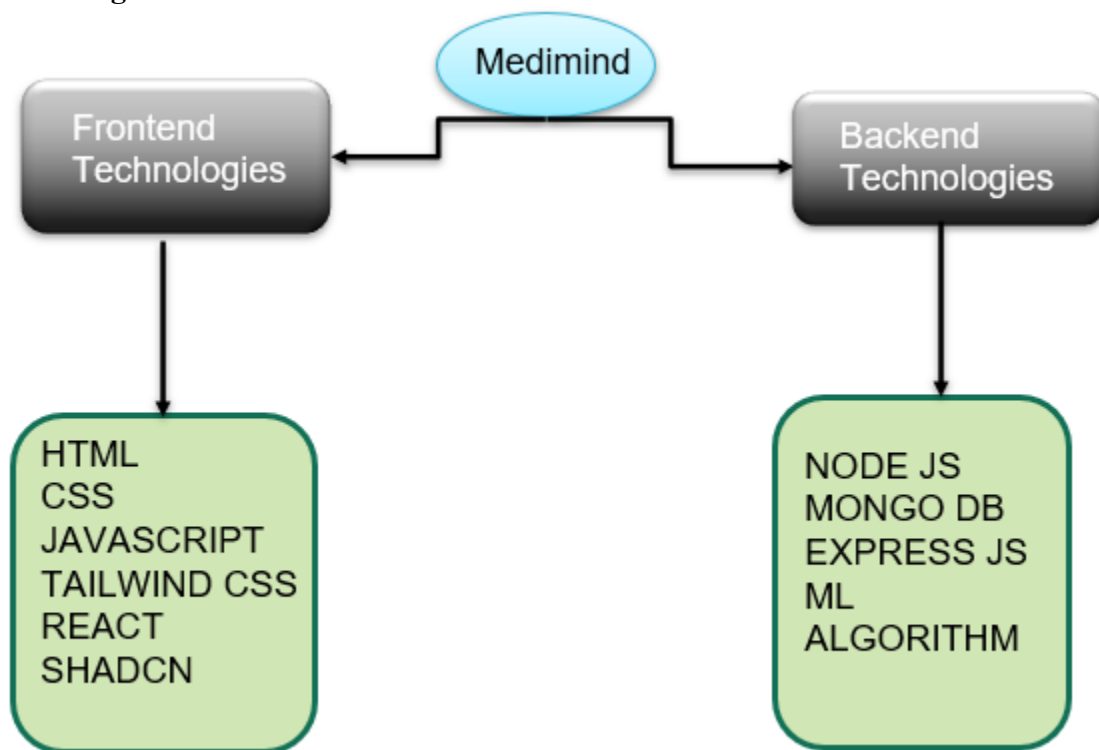
Functionalities of Proposed System

Chapter 3: Functionalities of Proposed System

3.1 Functionalities

- 4 Multi-Disease Prediction: The system is designed for multi-disease health analysis, specifically covering heart disease, diabetes, breast cancer, and lung cancer
- 5 Secure User Authentication: The system provides a secure login facility to manage user access and protect personal data
- 6 Multi-Modal Data Input: Users can upload various forms of medical data, including prescriptions and medical images like X-rays or CT scans.
- 7 AI-Powered Analysis: The system uses different machine learning algorithms (Logistic Regression, Decision Tree, and Naive Bayes) to provide accurate predictions
- 8 Custom PDF Report Generation: Users have the ability to download their analysis results in a custom PDF report for easy access and record-keeping.
- 9 User Data Management: The platform uses MongoDB to securely store user details and their historical analysis records, creating a personal health timeline.

9.1 Flow Diagram

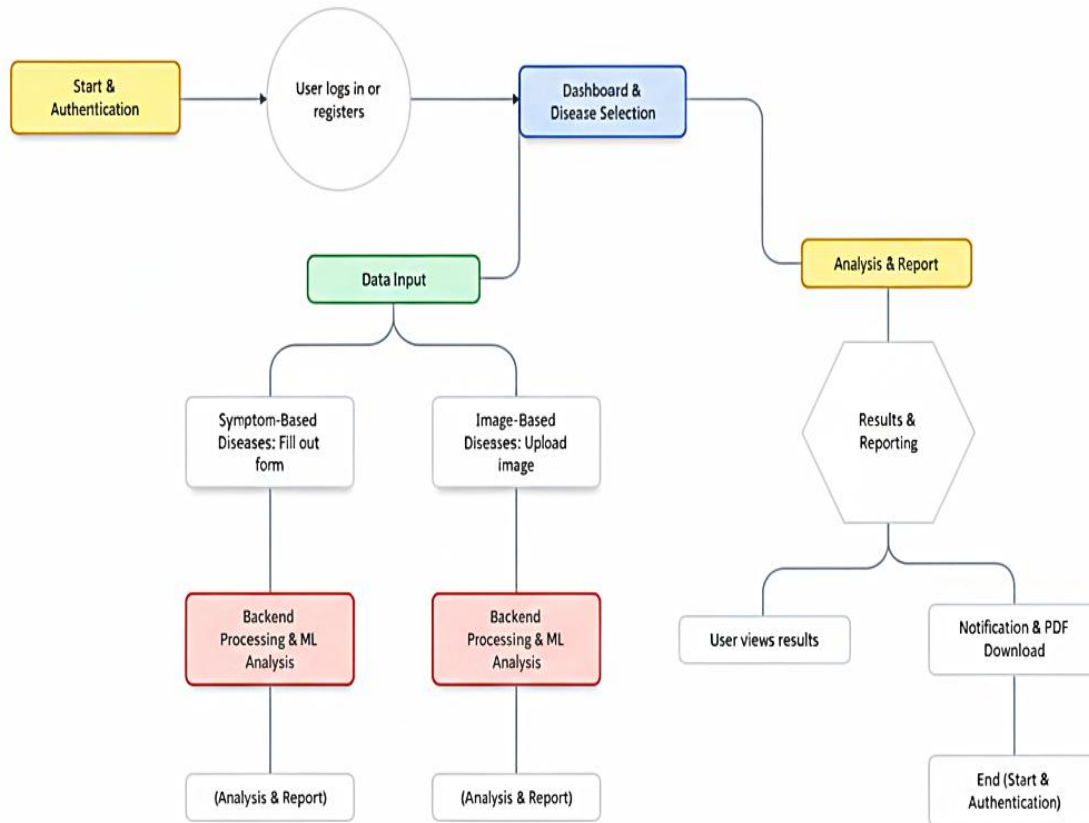


Chapter 4:

IMPLEMENTATION

Chapter 4: IMPLEMENTATION

4.1 Implementation



4.2 Description

The proposed system, MediMind, is a web-based intelligent health prediction platform that utilizes machine learning algorithms such as Logistic Regression, Decision Tree, and Naïve Bayes to detect and predict multiple diseases including Heart Disease, Diabetes, Breast Cancer, and Lung Cancer. The system allows users to securely upload their medical reports or diagnostic images and instantly receive accurate predictions about their health condition along with a downloadable report. Unlike existing manual systems, MediMind provides an automated, multi-disease, and user-friendly solution that helps in early diagnosis and timely intervention. It is developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js) which ensures high scalability, data security, and fast performance. All user data is stored safely in MongoDB, maintaining privacy and confidentiality through encryption. This system bridges the gap between technology and healthcare by providing users with an accessible, accurate, and secure means to monitor their health from anywhere, thereby promoting preventive healthcare and reducing dependency on manual diagnosis.

4.3 Code

4.3.1 Data Collection:

```
import { spawn } from "child_process";
import { fileURLToPath } from "url";
import { dirname, resolve } from "path";
import path from "path";
import fs from "fs";
import multer from "multer";
import { asyncHandler } from "../utils/asyncHandler.js";
import { ApiError } from "../utils/ApiError.js";

// --- Multer Setup ---
const __filename = fileURLToPath(import.meta.url);
const __dirname = dirname(__filename);

const storage = multer.diskStorage({
  destination: (req, file, cb) => {
    cb(null, path.join(__dirname, "../uploads"));
  },
  filename: (req, file, cb) => {
    cb(null, `${Date.now()}-${file.originalname}`);
  },
});
export const upload = multer({ storage: storage });

// --- Helper for Deleting Files ---
const deleteFile = (filePath) => {
  fs.unlink(filePath, (err) => {
    if (err) {
      console.error(`Error deleting file: ${filePath}`, err);
    }
  });
};

// --- Define the absolute path to the Python in your venv ---
const pythonExecutable = resolve(__dirname, "..", "venv", "Scripts", "python.exe")
```

4.3.2 Training model:

```
import os
import sys
import numpy as np
import tensorflow as tf
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing import image
from tensorflow.keras.utils import custom_object_scope

# Suppress TensorFlow INFO and WARNING logs
os.environ["TF_CPP_MIN_LOG_LEVEL"] = '2'
```

```
# The "blueprint" for the custom layer, now with robust input handling
class CustomScaleLayer(tf.keras.layers.Layer):
    def __init__(self, scale=1./255, **kwargs):
        super(CustomScaleLayer, self).__init__(**kwargs)
        self.scale = scale/18
```

4.3.3 Testing model:

```
const breastpred = asyncHandler(async (req, res) => {
  if (!req.file) throw new ApiError(400, "No image file uploaded");
  const filePath = req.file.path;
  const scriptPath = resolve(__dirname, "../ML/Breast Cancer Prediction/breast_cancer_prediction.py");

  const pythonProcess = spawn(pythonExecutable, [scriptPath, filePath]);

  let predictionData = "";
  let errorData = "";

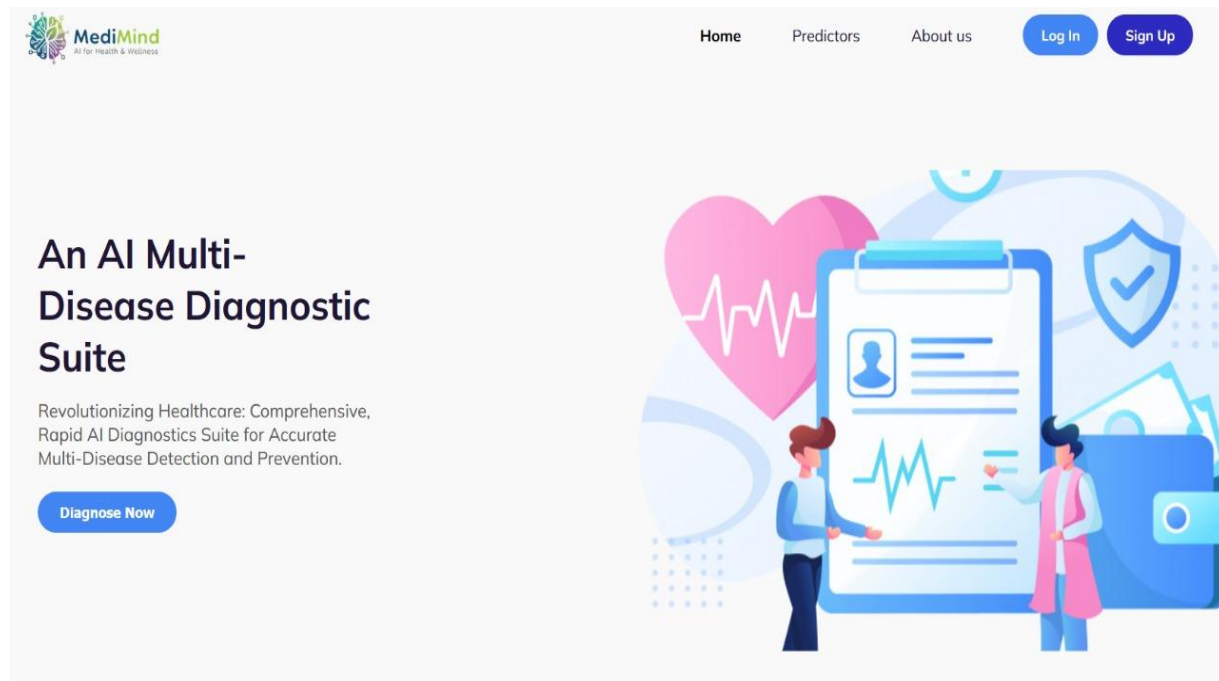
  pythonProcess.stdout.on("data", (data) => (predictionData += data.toString()));
  pythonProcess.stderr.on("data", (data) => (errorData += data.toString()));

  pythonProcess.on("close", (code) => {
    deleteFile(filePath); // Always delete the file
    if (code !== 0) {
      console.error("Breast prediction stderr:", errorData);
      return res.status(500).json({ error: "Prediction script failed", details: errorData });
    }

    console.log("Raw output from Python script (Breast):", predictionData.trim());

    return res.status(200).json({ prediction: predictionData.trim() });
  });
});
export { heartpred, diabetespred, lungpred, breastpred };
```

RESULT :



The Comprehensive Health Diagnostics Suite utilizes advanced AI technology for early detection and precise prediction...



Heart Disease

Guarding Hearts: AI solutions for accurate prediction and early intervention in heart disease.



Lung Cancer

Clearing the Air: AI-driven insights for proactive lung cancer prediction and care.



Breast Cancer

Beyond Detection: AI innovation for early, precise breast cancer prediction and care.



Diabetes

Empowering Health: AI solutions for precise diabetes prediction and proactive wellness.

DIABETES PREDICTOR

Don't want to type manually? Upload your report and we will do it for you

[Upload Report](#) [Test Reports](#)

[Predict](#)

DIABETES PREDICTOR

Prediction Result:

Number of Pregnancies: 1
Glucose Level: 85
Blood Pressure: 66
Skin Thickness: 29
Insulin: 0
BMI (Body Index Mass): 26.6
Diabetes Pedigree Function: 0.351
Age: 31

The person is not suffering from Diabetes

[Download Report](#)

[Re-predict](#)

Chapter 5:
CONCLUSION AND
FUTURE WORK

Chapter 5

Conclusion and Future Plan of Work

The proposed system, MediMind, provides an intelligent and efficient solution for early detection of critical diseases such as Heart Disease, Diabetes, Breast Cancer, and Lung Cancer using machine learning algorithms. It offers users a convenient platform to upload their medical reports and receive instant, accurate, and reliable predictions, helping them take preventive measures at the right time. By integrating technologies like MongoDB for secure data storage and the MERN stack for smooth performance, the system ensures scalability, privacy, and ease of use. MediMind successfully bridges the gap between healthcare and technology, empowering individuals to monitor their health from anywhere and promoting early diagnosis through data-driven insights.

In the future, this system can be further enhanced by integrating deep learning models such as Convolutional Neural Networks (CNNs) for more precise image-based disease detection, and expanding its scope to include additional diseases. Features like real-time doctor consultation, IoT-based health monitoring, and mobile application support can also be implemented to make the system more interactive and accessible. Incorporating blockchain technology in future versions could further strengthen data security and trust, ensuring complete transparency in medical data handling and storage.

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